

INTRODUCTION

Rivers are among the most biodiverse ecosystems on the planet. A river is a natural stream of water that flows from a high source to another body of water and are crucial to human survival by providing potable water to people globally. In the UK, this is evident as roughly 66% of drinking water stems from rivers, lakes and reservoirs.

Lakes and rivers require low levels of nutrients (such as NO_3^- and PO_4^{3-}) and pollutants to support wildlife, so we have a role to play in preventing the bioaccumulation of these substances – both to provide potable water worldwide and to allow rivers to remain a viable habitat for a wide range of animals and organisms.

Rivers also carry foul water away from our homes, moving sewage via combined sewer overflow systems and helping prevent sewage backup during heavy rainfall, which keeps our streets clean. These points motivate our study, in which we investigated the upstream, downstream and discharge concentrations of algae, bacteria, NO_3^- and PO_4^{3-} . High concentrations of these indicate low water quality, conflicting with the societal interests outlined above.

METHODS

Water samples were collected from three locations along the river: upstream, downstream and at the point of discharge.

1. Algal count (microscopy)

- Concentrate the algae sample
- Quantify algae using a haemocytometer
- Multiply the count by 1000 to determine total algae in the original sample

2. Nutrient concentration (spectrophotometry)

- Prepare upstream, downstream and discharge samples
- Measure absorbance of all samples at 840 nm
- Construct and read from a calibration curve to determine phosphate and nitrate concentration

3. Bacterial count (spread plates)

- Prepare serial dilutions of the three samples
- Spread diluted samples on agar plates, and E. coli samples on EMB agar plates
- Count bacterial colonies to calculate bacteria count in the original sample

4. Antibiotic resistance (PCR & gel electrophoresis)

- Set up PCR to screen for antibiotic-resistance genes
- Run gel electrophoresis on an agarose gel
- Photograph and analyse the gel

RESULTS

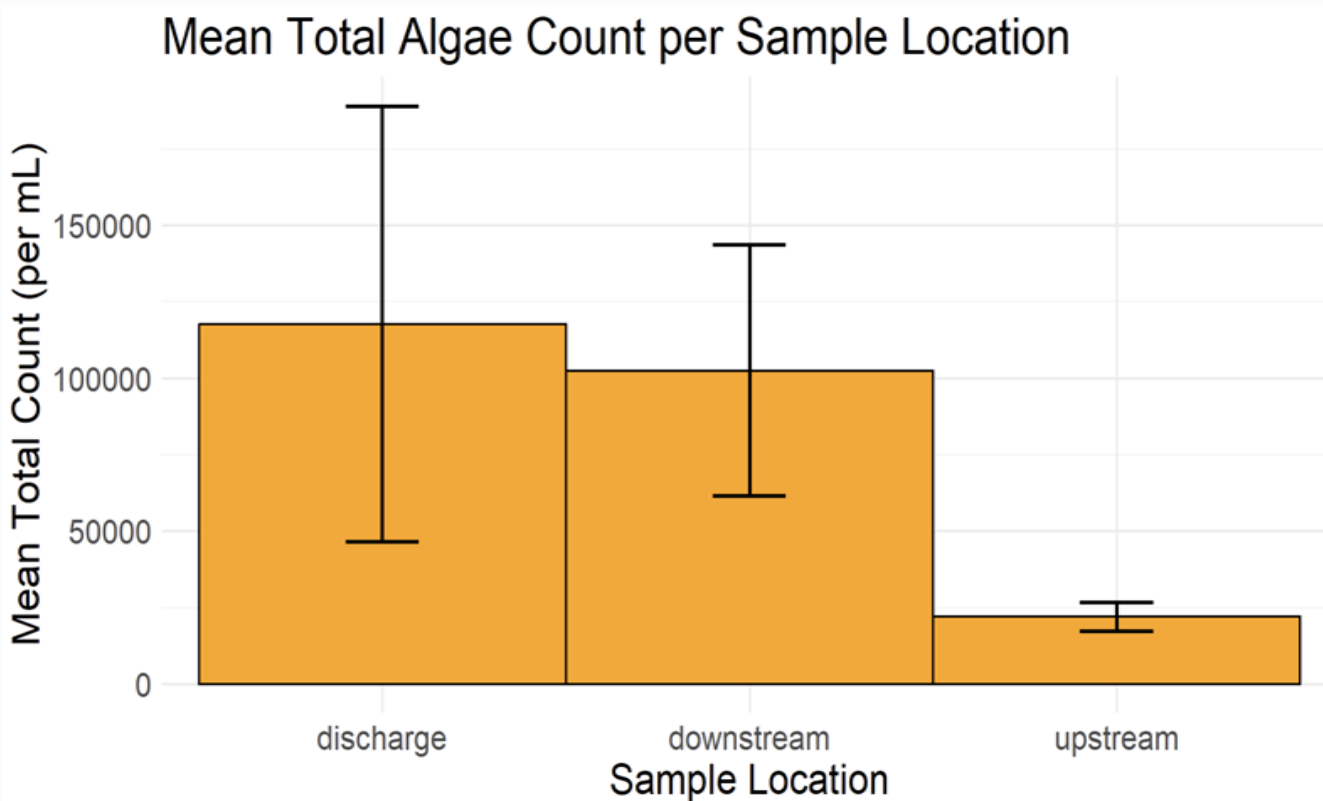


Figure 1. Mean total algal count per sample location, determined by microscopy and haemocytometer counts. Error bars show $\pm$ standard deviation.

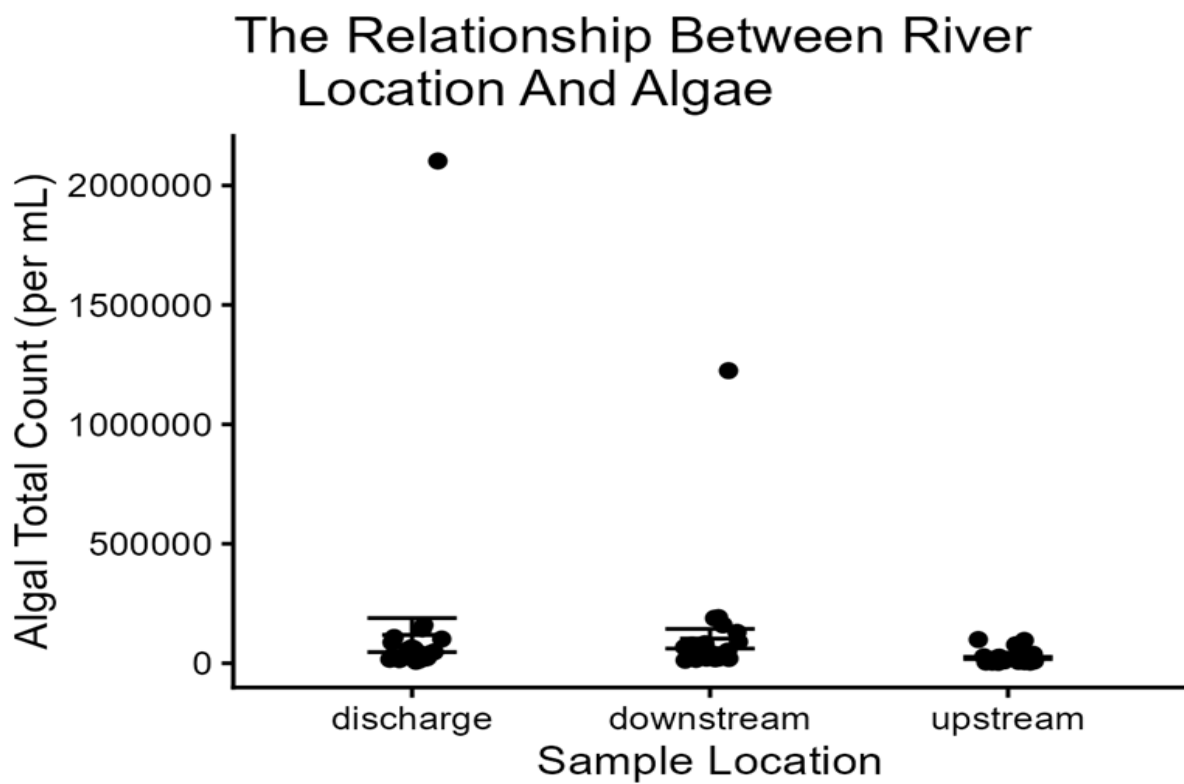


Figure 2. Relationship between river location and total algal count per sample.

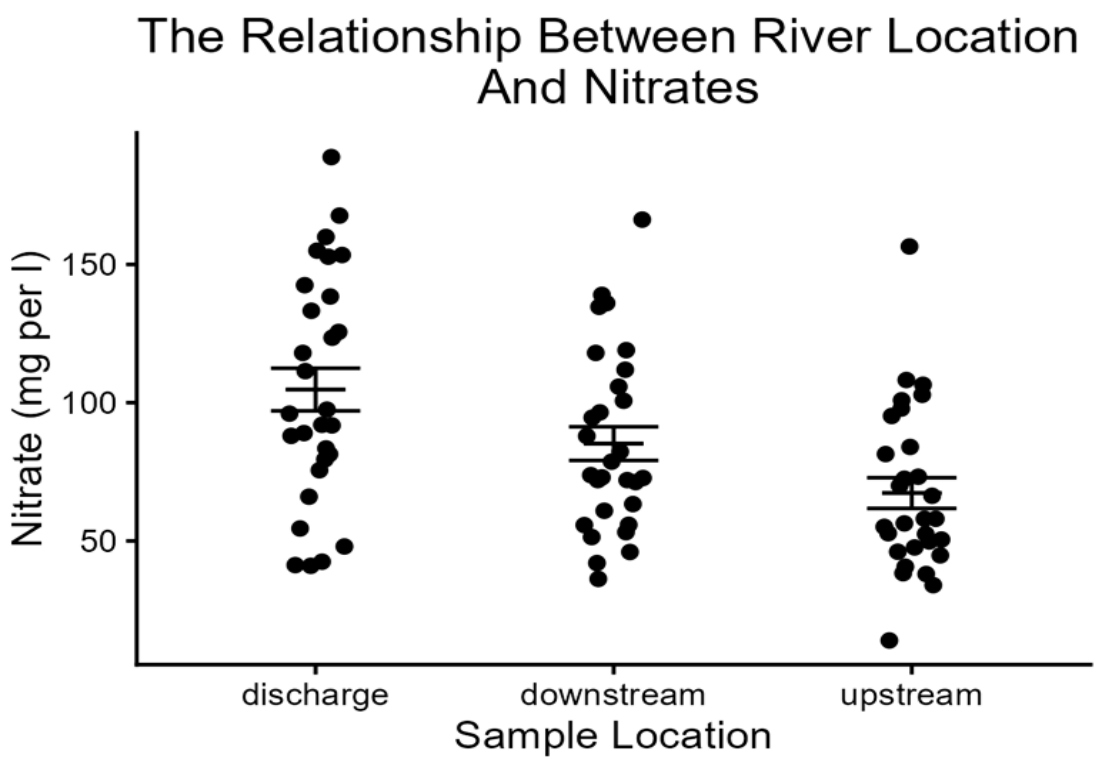


Figure 3. Relationship between river location and nitrate concentration (mg/l).

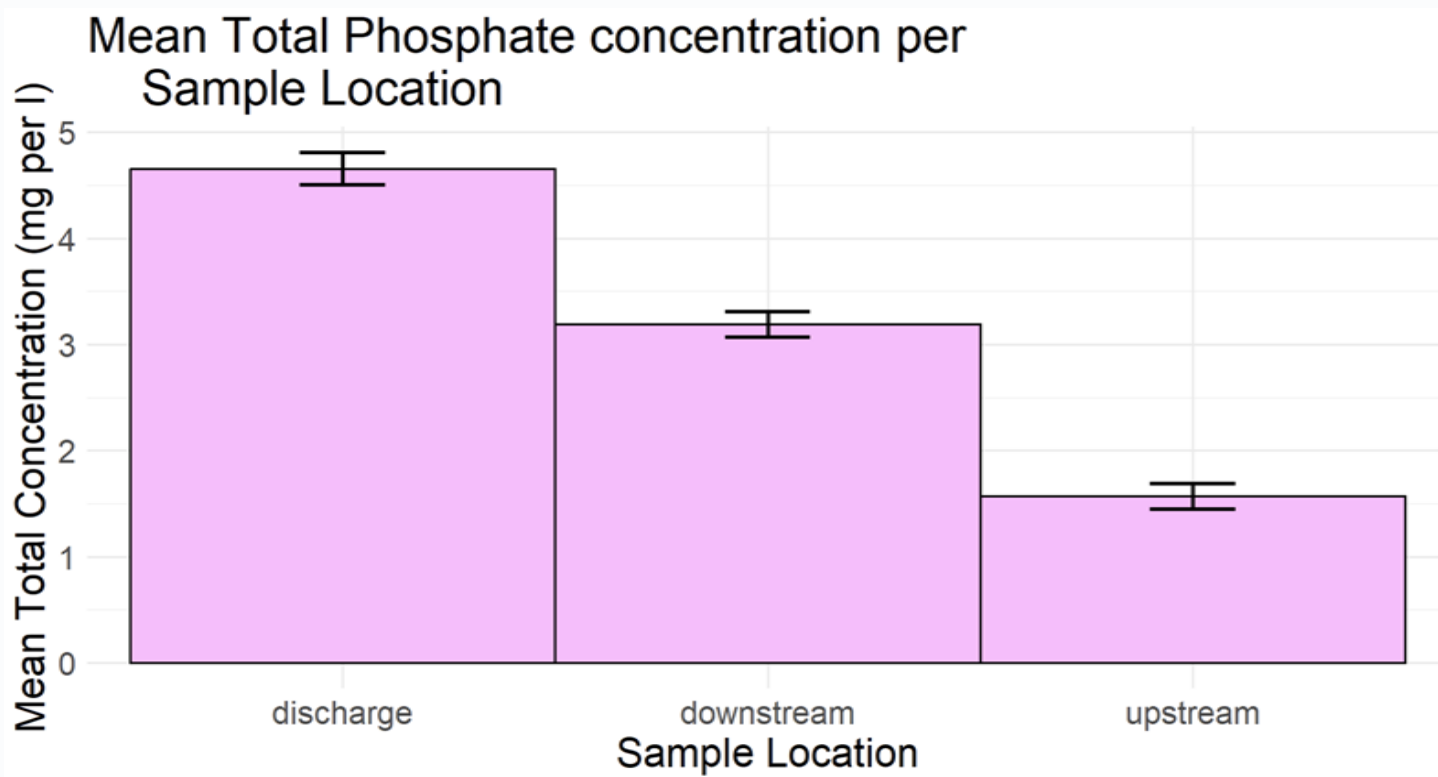


Figure 4. Mean total phosphate concentration (mg/l) per sample location. Error bars show $\pm$ standard deviation.

DISCUSSION

The data provide insight into river water quality along the upstream-to-discharge gradient. The general trend indicates that the discharge point is the most heavily impacted area of the river.

Algal counts were elevated above levels typically associated with clean water throughout the river, suggesting algal bloom – the rapid proliferation of algae, which has the potential to contaminate water and cause illness in humans. However, this result should be interpreted with some caution, as the counting method relied on manual identification and is therefore susceptible to low precision.

Phosphate and nitrate concentrations were high across all sample sites, particularly at the discharge and downstream locations. Both nutrients are established drivers of eutrophication – excessive algal growth caused by nutrient enrichment – which depletes dissolved oxygen and can produce toxins harmful to both the ecosystem and human health. As algae die, oxygen is consumed during decomposition, reducing the oxygen available to other aquatic organisms and consequently harming river biodiversity. Owing to human activity and the natural flow of water, pollutants are transported from upstream sources towards the downstream and discharge points; upstream sites, receiving less contamination, more closely reflect natural nutrient levels.

Elevated downstream and discharge nutrient levels indicate probable sewage or agricultural contamination, as plants absorb only 40-60% of applied fertilizer while the remainder leaches into water bodies.

The absence of antibiotic-resistant bacteria across all sample regions is encouraging, given the public health risks to recreational river users.

Overall, routine monitoring is vital to safeguard public health and environmental standards. Future work should expand sample sizes across seasons and identify specific eutrophication-linked algae such as cyanobacteria.

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